



Collaborative Cocaine Treatment Study: Longitudinal Analysis of Treatment Effect on Addiction Severity Index Legal Subscale

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Source: *Psychosocial Treatments for Cocaine Dependence: National Institute on Drug Abuse Collaborative Cocaine Treatment Study*

Background

- The **NIDA Collaborative Cocaine Treatment Study** randomly assigned one of four psychosocial treatments to 487 cocaine-dependent patients across five treatment centers. The active treatment period spanned six months.
- The **Addiction Severity Index (ASI)** is among the most widely used tools for assessing problems associated with substance abuse disorder.*
- The **Legal Subscale** of the ASI measures a patient's involvement with the legal system on a scale of 0 to 1, with 0 representing no reported legal problems and 1 representing severe legal problems.

**Source: J Subst Abuse Treat. 2013 Jul 23;45(5):444–450. doi: [10.1016/j.jsat.2013.06.003](https://doi.org/10.1016/j.jsat.2013.06.003)*

Questions

Is there a treatment effect on patient legal issues over the longitudinal period?

Is the treatment effect modified by whether patients completed the study?

Distribution of Legal Subscale at Baseline

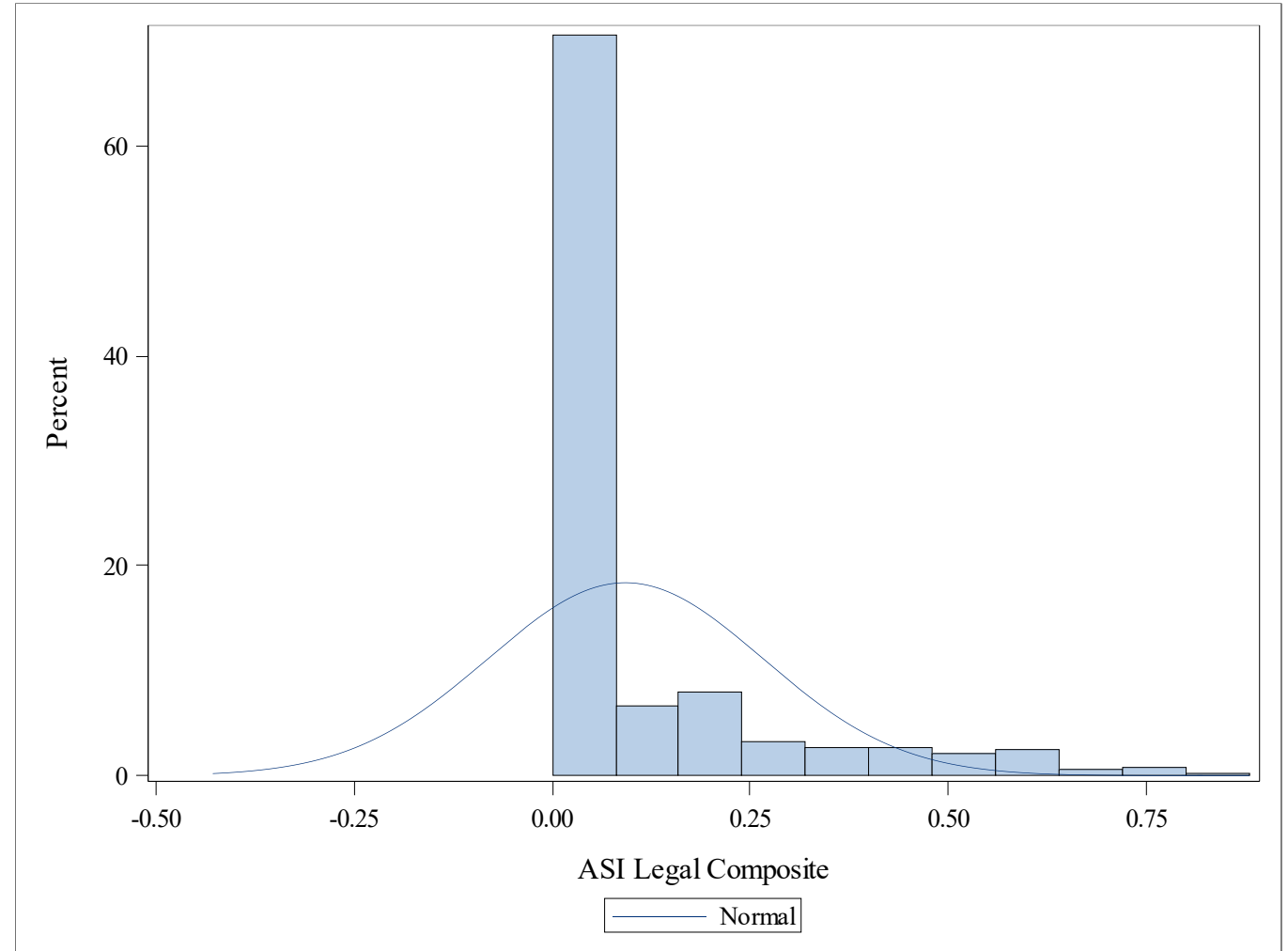
Median = 0

Mean = 0.09

75th Percentile = 0.12

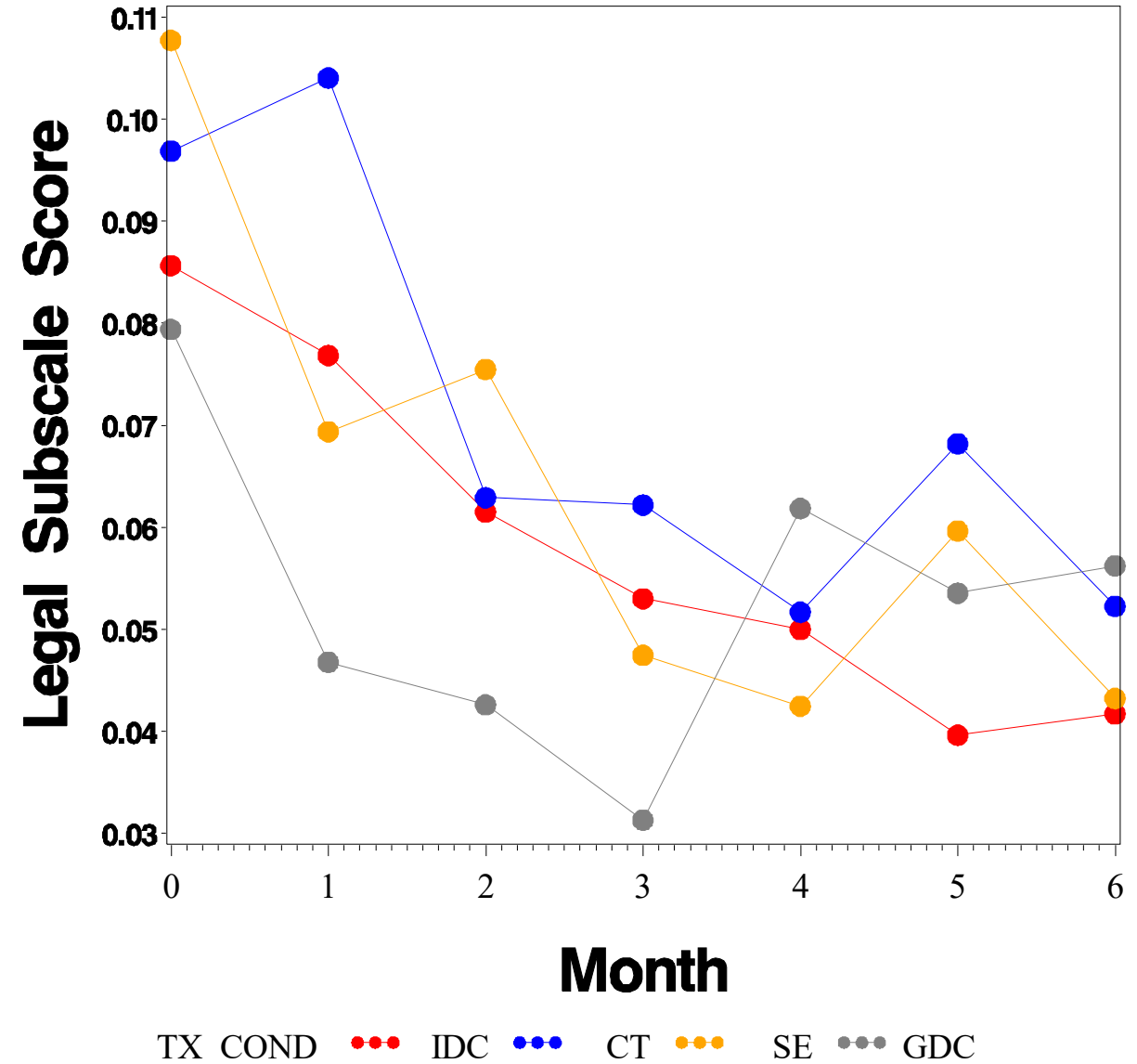
Maximum = 0.85

335 patients had a score of 0 at baseline (68.79%), and 152 patients had a score greater than 0 (31.21%).

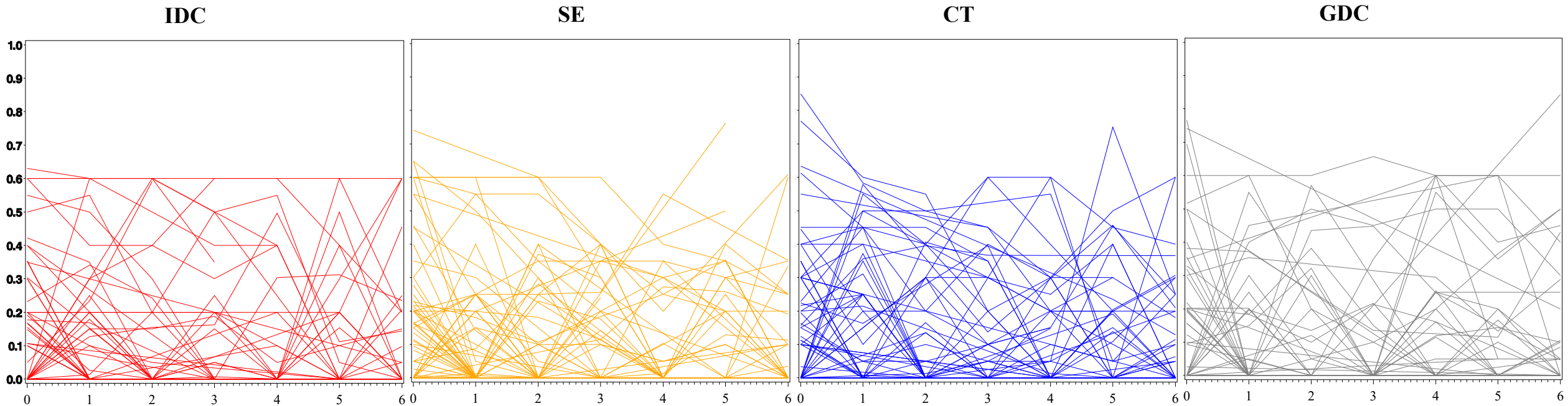


Mean Legal Subscale Scores by Month and Treatment Group

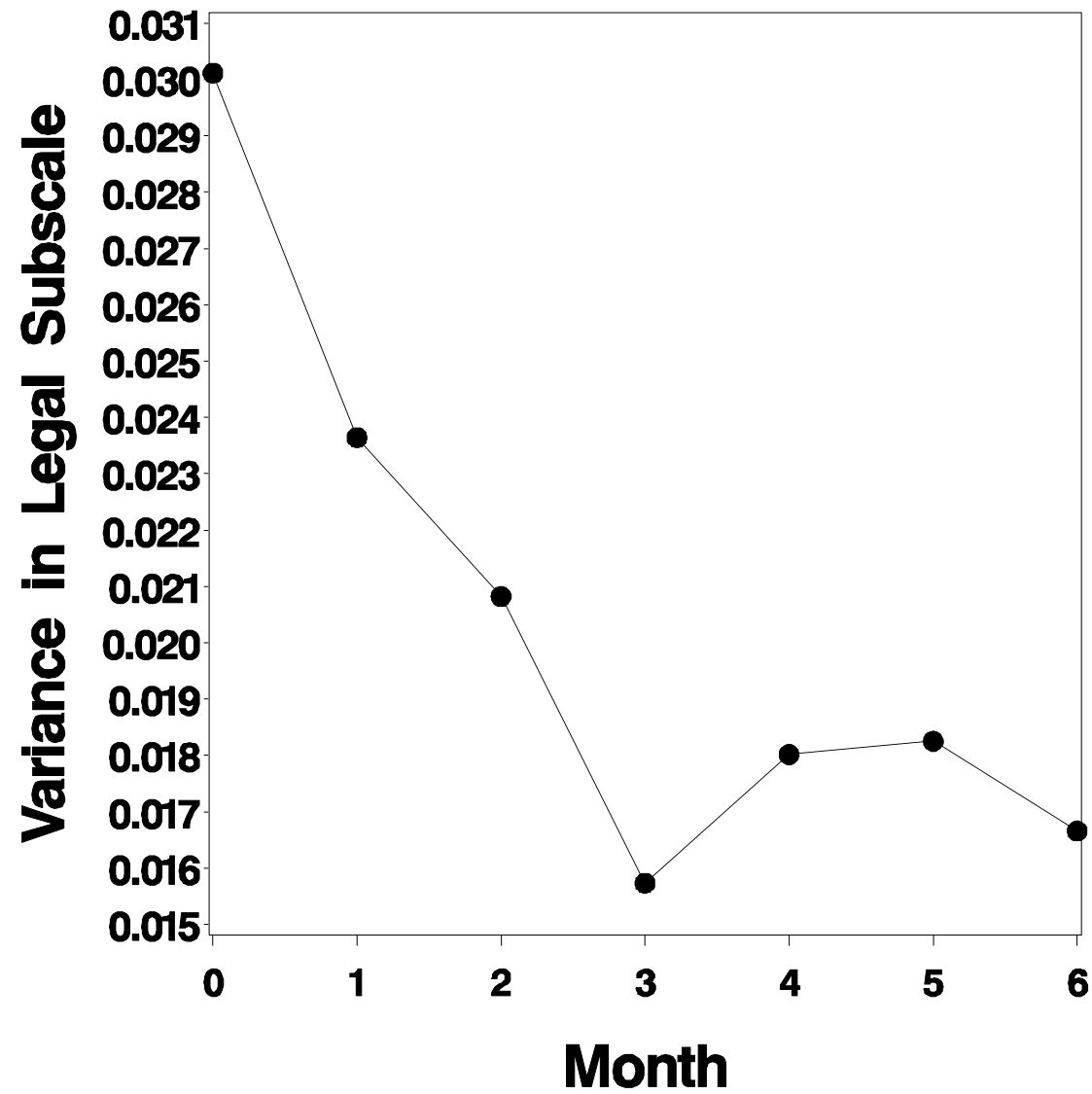
- Individual drug counseling plus group drug counseling (**IDC**)
- Cognitive therapy plus group drug counseling (**CT**)
- Supportive-expressive therapy plus group drug counseling (**SE**)
- Group drug counseling alone (**GDC**)



Individual Patient Legal Subscale Scores Over the Six-Month Period



Legal Subscale Variance by Month



Mixed Model ANOVA

- Treat time as categorical
- Include baseline as a covariate in the model

Question:

What is the best variance-covariance matrix?

Best Mathematical Form of Time

Form	Parameters	-2 Restricted Log Likelihood
Linear	4	-3398.02488936
Log Linear	4	-3410.39218238
Square Root	4	-3410.29157859
Quadratic	6	-3357.31946469
Cubic	8	-3313.31157566
Piecewise at Month 2	6	-3376.66129845
Piecewise at Month 3	6	-3371.66252197
Categorical	4	-3474.40249310

Best variance-covariance matrix

Covariance Structure	Parameters	-2 Restricted Log Likelihood
Compound Symmetry	2	-3165.17148713
Autoregressive	2	-3375.42286077
Toeplitz	6	-3386.74805683
Unstructured	21	-3474.40249310

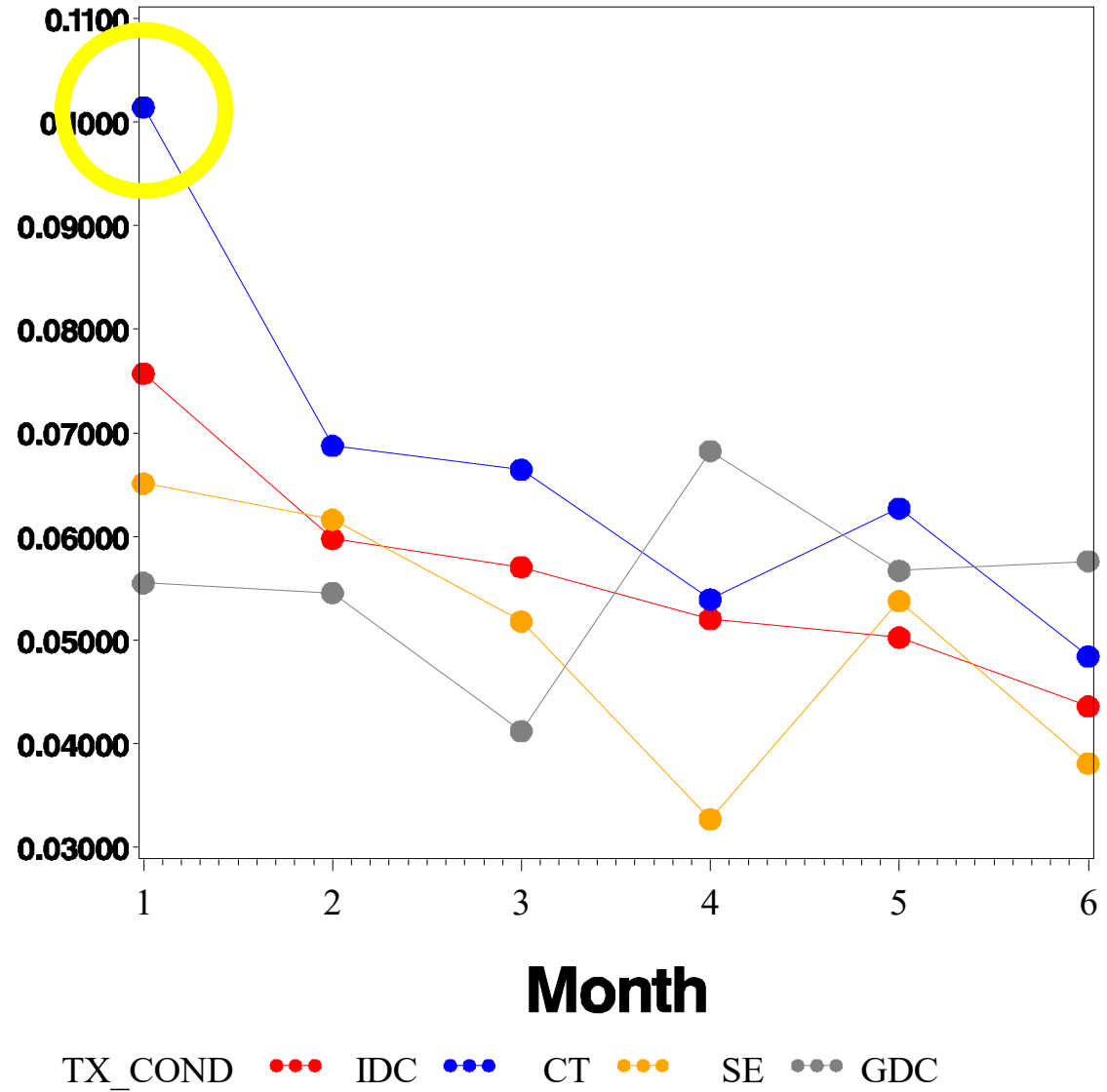
Unstructured has a significant gain in fit compared to Toeplitz ($p < 0.0001$).

Final Mixed Model ANOVA Results

- Baseline is highly correlated with legal subscale ($p < 0.0001$).
- There is not sufficient evidence of a treatment main effect ($p = 0.5835$).
- There is not sufficient evidence that legal subscale scores vary differentially between treatment groups across the six months ($p = 0.2766$).
- There is a significant main effect of time ($p = 0.0137$).

MMANOVA Model-Based Estimates

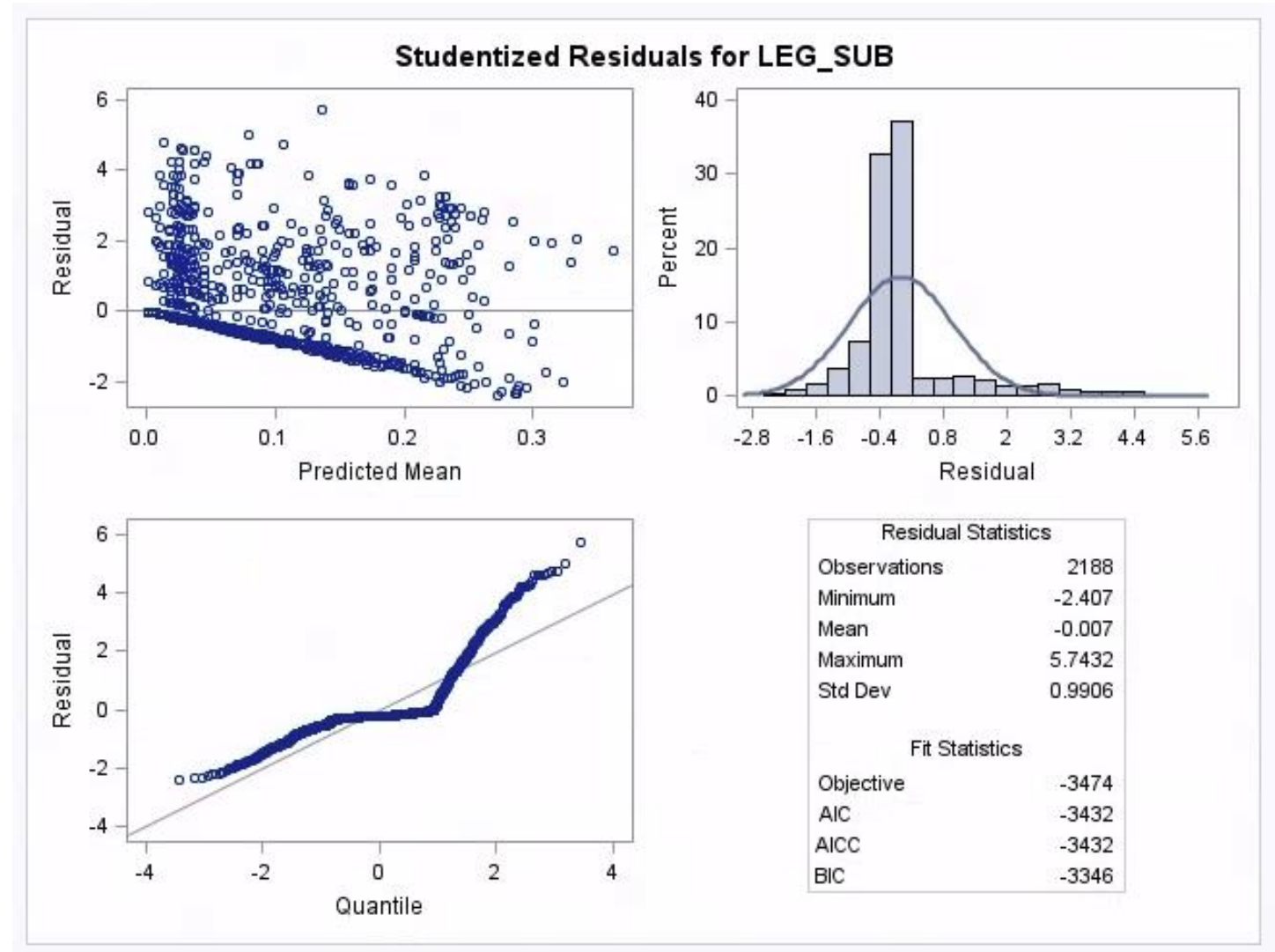
Average Legal Subscale
Score by Treatment Group
and Month



MMANOVA

Goodness of Fit

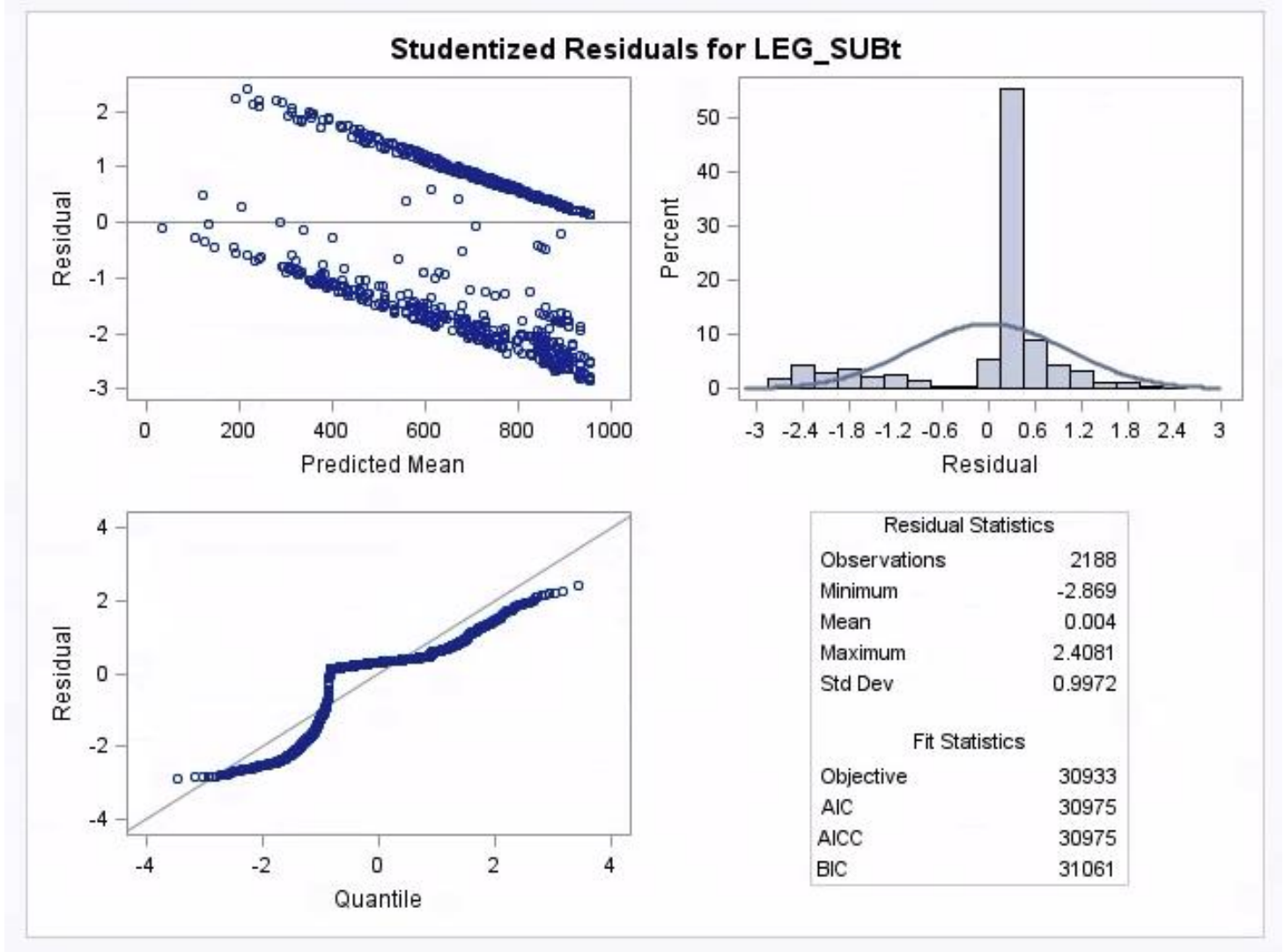
Studentized Residuals from the model are not normally distributed (Kolmogorov-Smirnov $p=0.01$).



MMANOVA Box Cox Procedure

- The Box Cox procedure suggests a best lambda of -3.00.
- However, transforming the outcome does not normalize the model residuals.

What about modeling the outcome as count or binary?



Generalized Linear Mixed Model

- Outcome is not normally distributed.
- There is no clear linear form of time; fit time as a category.
- Include baseline Legal Subscale scores as a covariate.

Questions:

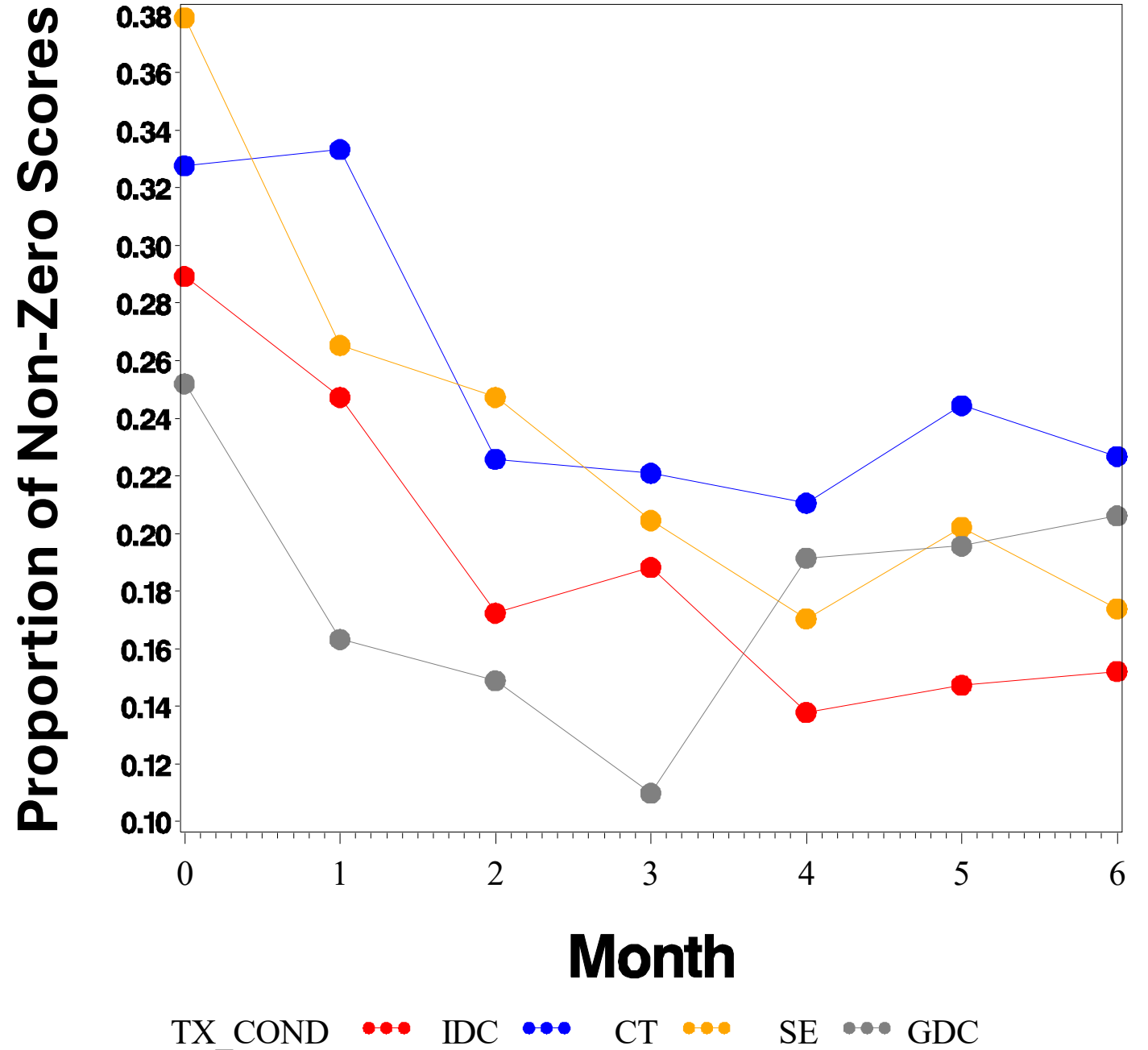
- What is the best form of the outcome?
- What is the best residual covariance structure?

Best form of the outcome?

Outcome	Generalized Chi-Square/DF
Continuous	0.01
Binary (Binomial)	0.60
Count (Poisson)	0.18
Negative Binomial	[Did not converge]

Binary Outcome: Legal Issues

Legal subscale = 0 or
Legal subscale > 0



Best residual covariance structure?

Covariance Structure	Parameters	-2 Restricted Log Pseudo-Likelihood
Compound Symmetry	2	10307
Autoregressive	2	10317
Toeplitz	6	[Did not converge]
Unstructured	21	[Did not converge]

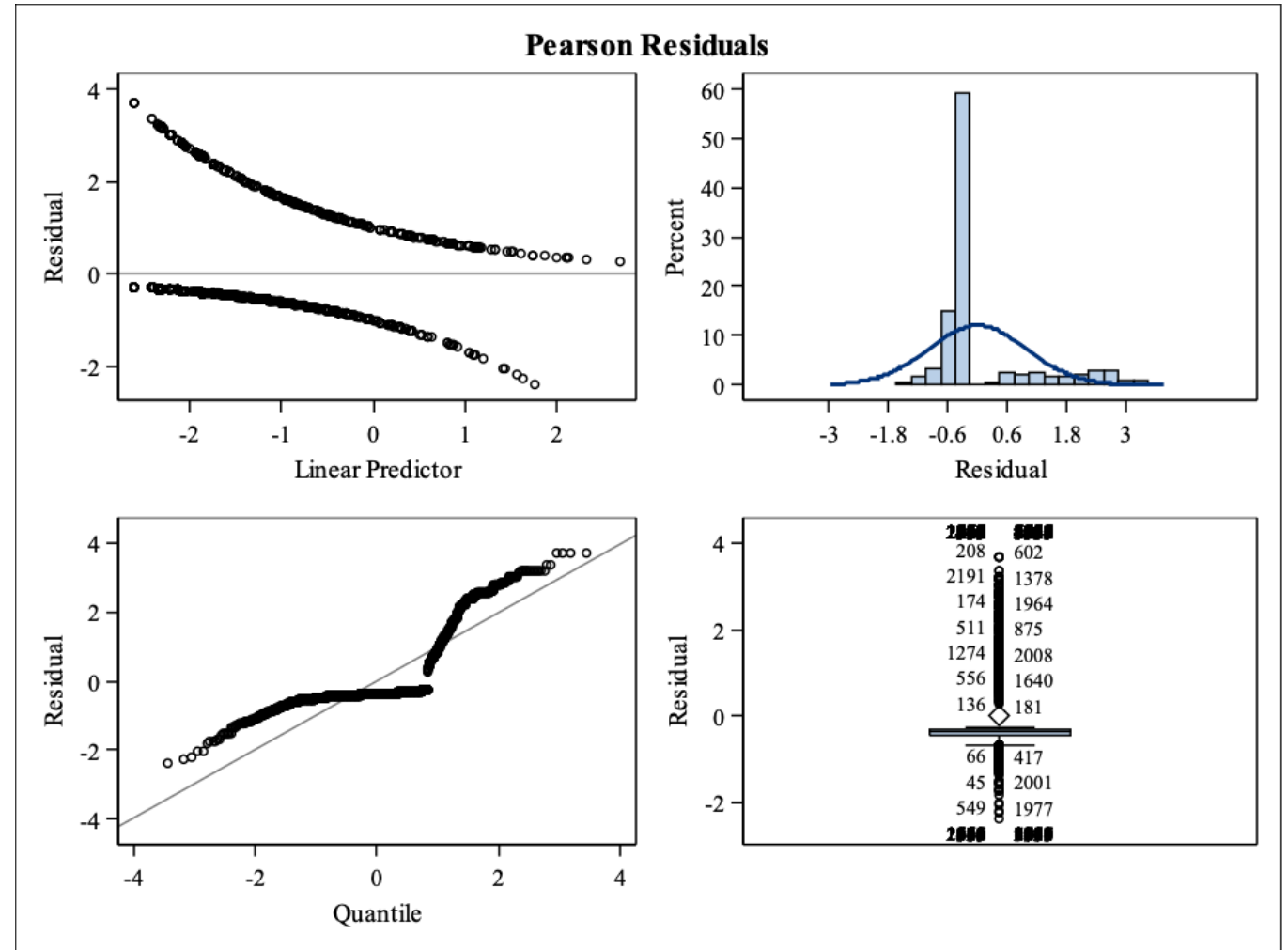
The final model fits a residual matrix with a compound symmetry design to account for within-subject correlation over time.

Final Generalized Linear Mixed Model Results

- Baseline legal subscale scores are highly correlated with whether a patient has legal problems over the six-month period ($p < 0.0001$).
- There is not sufficient evidence of a treatment main effect on binary legal issues ($p = 0.2502$).
- There is not sufficient evidence of a differential treatment effect on binary legal issues over the longitudinal period ($p = 0.4412$).
- There is a significant main effect of time ($p = 0.0103$).

Goodness of Fit

- The residuals are not normally distributed (Kolmogorov-Smirnov p-value=0.01).
- The distribution shape suggests issues with zero-inflated outcome.



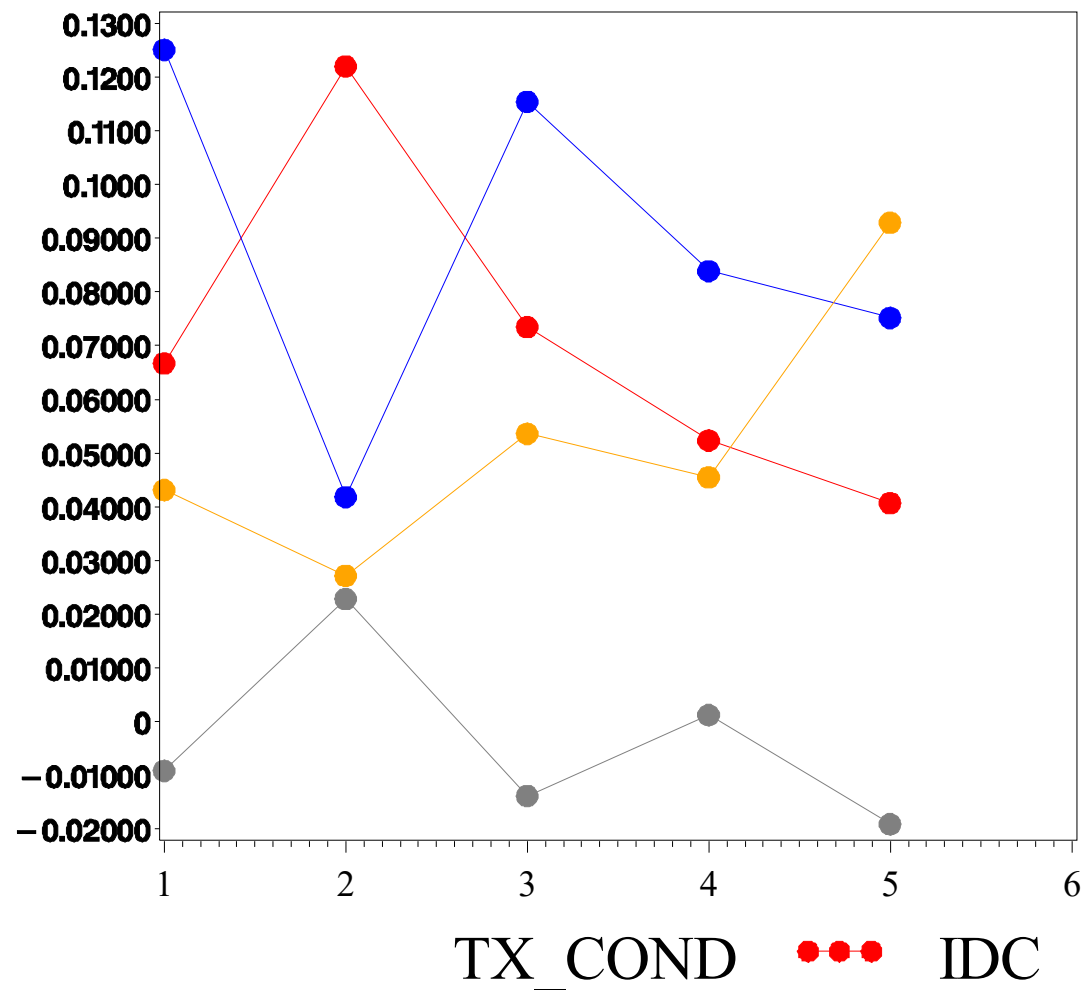
Completers vs. Dropouts

- 335 patients began the study with no legal issues.
 - 71 dropped out (21.19%)
 - 264 completed the study (78.81%)
- 152 patients began the study with legal issues.
 - 38 dropped out (25%)
 - 114 completed the study (75%)

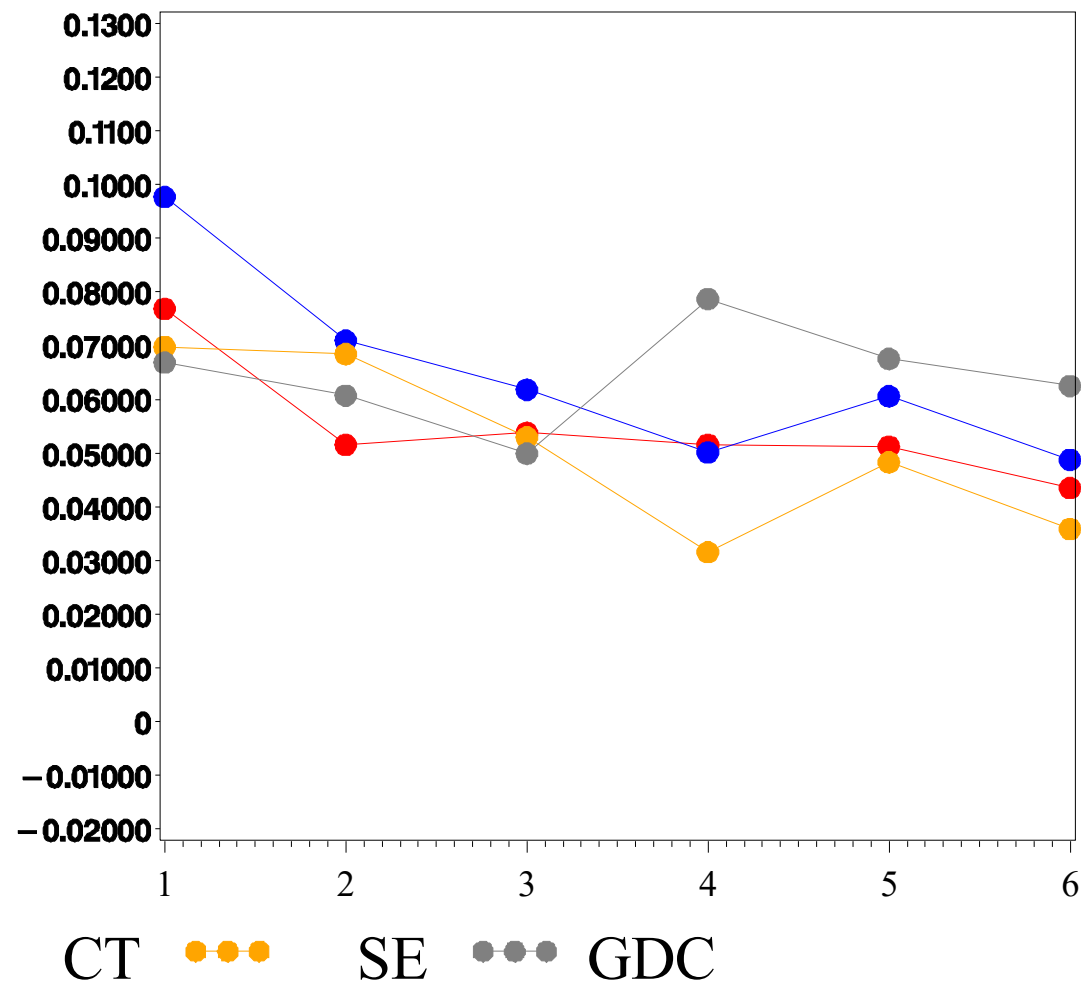
Are missing legal subscale scores biasing the estimated effects over the longitudinal period?

Pattern Mixture Model Results

Dropout



Complete



Summary

- Most patients in the study had no legal issues at baseline.
- Baseline ASI Legal Subscale scores were highly predictive of patients' legal issues over the study period.
- Patient legal issues did not vary differentially between treatment groups over the study period.
- Patient legal issues appeared to decline overall over the course of the study; however, the missing-data pattern may be informative.